Cache Design 2

CS 2410

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Cache Design Parameter 5: Write-Through vs. Write-Back

Writes and Cache Consistency

- Assume **&x** is 111010_2, and x == 24 initially

```
lw   t0, &x
addi t0, t0, 1        # x++
sw   t0, &x
```

- How will the **lw** change the cache?
- How will the **sw** change the cache?
  - Uh oh, now the cache is **inconsistent**. (Memory still has the old value 24.)

	V	Tag	Data
000	0		
001	0		
010	1	111	25
011	0		
100	0		
101	0		
110	0		
111	0		

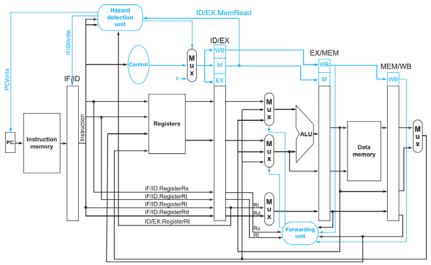
- How can we solve this? Two policies:
  - **Write-through**: Propagate write all the way through memory
  - **Write-back**: Write back cache block when it is evicted from cache

Write-Through Policy

Policy 1: Write-through

- Write-through:
  - On hit, write to cache block and **propagate** write to lower memory
  - On miss, keep on propagating the write to lower memory
- What happens if we write **25** to address **111010_2**?
- What happens if we write **94** to address **000010_2**?

→ Caches are kept consistent at all points in time!



L1 Cache

	V	Tag	Data
000	0		
001	0		
010	1	111	25
...		...	...

L2 Cache

	V	Tag	Data
000	0		
001	0		
010	1	000	94
...		...	...

Memory

Address	Data
...	...
000010	94
...	...
111010	25
...	...

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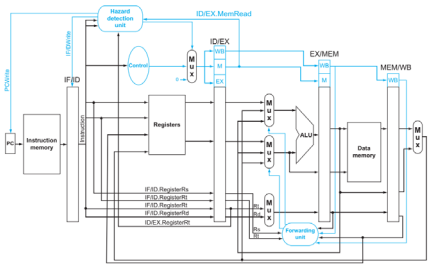
Write-through: Allocate reads but not writes

- What happens if we read from address **000010**$_2$?
  - We can just discard the conflicting cache block **111010**$_2$
  - It's just an extra copy of the same data
- Note how we allocate blocks only on read misses
  - Write misses don't allocate blocks because it doesn't help anyway --- writes are propagated to lower memory even on write hits
  - This policy is called **no write allocate**



L1 Cache

	V	Tag	Data
000	0		
001	0		
010	1	000	94
...		...	...

L2 Cache

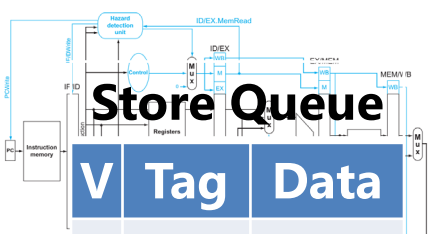
	V	Tag	Data
000	0		
001	0		
010	1	000	94
...		...	...

Memory

Address	Data
...	...
000010	94
...	...
111010	25
...	...

Write-through: Drawbacks

- Drawback: **Long write delays** regardless of hit or miss
  - Must always propagate writes all the way to DRAM
- Solution: **Store Queue** maintains pending writes
  - CPU gets on with work after moving pending write to store queue
  - But does the store queue solve all problems?



Store Queue

V	Tag	Data

L1 Cache

	V	Tag	Data
000	0		
001	0		
010	1	111	25
...		...	...

L2 Cache

	V	Tag	Data
000	0		
001	0		
010	1	000	94
...		...	...

Memory

Address	Data
...	...
000010	94
...	...
111010	25
...	...

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Write-through: Drawbacks

- The store queue does not solve all problems.

1. Store queue must be big enough to store all pending writes
   - May take more than 100 cycles for write to propagate to memory
   - Store queue always checked on every load → may add to **hit time**

2. Store queue buffer does not solve **bandwidth** problems
   - If memory bandwidth < rate of writes in program,

- Impractical to write-through all the way to memory
  - Typically only L1 caches are write-through, if any
- We need another strategy that is not so bandwidth-intensive

Write-Back Policy

Policy 2: Write-back

- **Dirty** block: a block that is temporarily inconsistent with memory
  - On a hit, write to cache block, marking it dirty. **No propagation**.
  - **Write back** dirty block to lower memory only when it is **evicted**
    → Saves bandwidth since write hits no longer access memory
- A **dirty bit** is added to the cache block metadata (marked "D")
  - Block **000001**$_2$ is clean → can be discarded on eviction
  - Block **111010**$_2$ is dirty → needs to be written back on eviction

Cache

	V	D	Tag	Data
000	0	0		
001	0	0	000	93
010	1	1	111	25
…			…	…

consistent

inconsistent

Memory

Address	Data
…	…
000001	93
…	…
111010	24
…	…

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Write-back: Allocate both reads and writes

- Unlike write-through, write hits do not result in propagation
  - May be beneficial to allocate writes to stop further misses!
  - Unlike write-through, write-back has a **write allocate** policy

- On write allocation, the block is **read** in from **lower memory**
  - Why the wasted effort?
  - Aren't we going to overwrite the block anyway with new data?
  - Why read in data that is going be overwritten?

Write-back: Write allocate

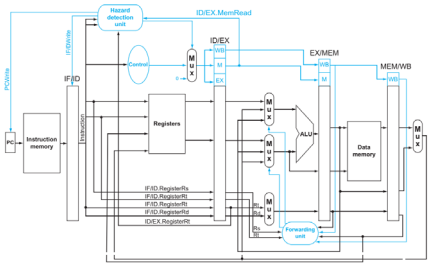
- Because a block is multiple bytes, and you are updating just a few
  - Suppose a cache block is 8 bytes (2 words)
  - Suppose you are writing to only the **first word**

V	D	Tag	Data	
1	1		**first word (written)**	**second word (not written)**

  - After allocate, the entire cache block is marked **valid**
    - That means **second word** as well as **first word** must be valid
    - That means **second word** must be fetched from lower memory
    - Otherwise if later **second word** is read, it will contain junk data
    - Unavoidable, unless you have a valid bit for each byte
      - That means spending 1 bit for every 8 bits of data
      - That's just too much metadata overhead

Write-back: Writes

- What happens if we write **25** to address **111010**$_2$?



Store Queue

V	Tag	Data

L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	111	24
...			...	...

L2 Cache

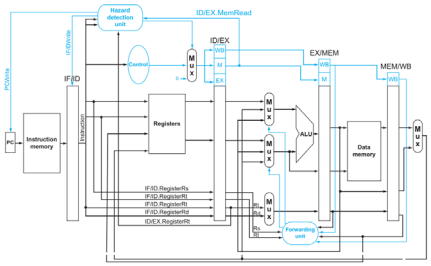
	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93
...			...	...

Memory

Address	Data
...	...
000010	93
...	...
111010	24
...	...

Write-back: Writes

- What happens if we write **25** to address **111010**$_2$?
  - L1 Cache **hit**! Update cache block and mark it dirty.
  - That's it! How quick is that compared to write-through?



Store Queue

V	Tag	Data

L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	111	25
...			...	...

L2 Cache

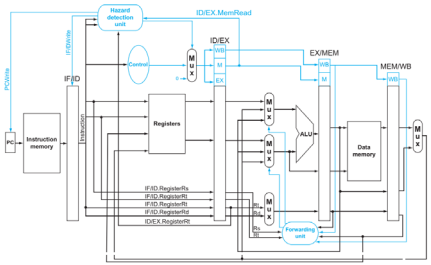
	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93
...			...	...

Memory

Address	Data
...	...
000010	93
...	...
111010	24
...	...

Write-back: Writes

- What happens if we write **94** to address **000010**$_2$?
  - L1 Cache **miss**! First thing we will do is add store to **Write Buffer**. (So that the CPU can continue executing past the store)



Store Queue

V	Tag	Data
1		94

L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	111	25
...			...	...

L2 Cache

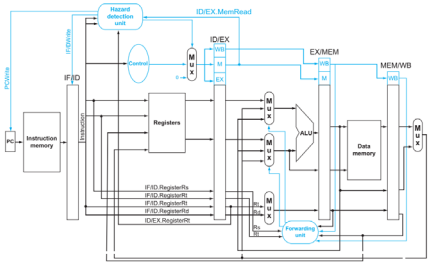
	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93
...			...	...

Memory

Address	Data
...	...
000010	93
...	...
111010	24
...	...

Write-back: Writes

- What happens if we write **94** to address **000010**$_2$? (cont'd)
  - Next the L2 Cache is searched and it's a **hit**!
  - To bring in block to L1 Cache, we first need to evict block **25**.
  - It's a dirty block, so we can't just discard it. Need to **write** it **back**!
  - Since block **25** misses in L2, it will take the **long** trip to Memory
  - Is there a way to put it aside and get to it later?



Store Queue

V	Tag	Data
1		94

L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	111	25
...			...	...

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93
...			...	...

Memory

Address	Data
...	...
000010	93
...	...
111010	24
...	...

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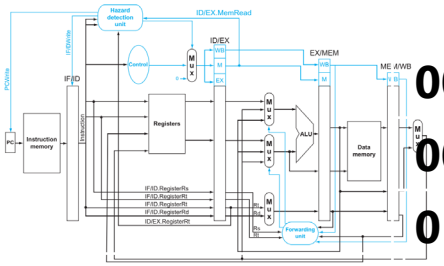
Write-back: Writes

- What happens if we write **94** to address **000010**$_2$? (cont'd)
  - Yes! Add **Write Buffer** to cache, just like store queue for pipeline!
  - Move block to L1 Write Buffer so L1 Cache can continue working
  - Pending block will get written back to Memory eventually



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	111	25

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93

Memory

Address	Data
...	...
000010	93
...	...
111010	24
...	...

Store Queue

V	Tag	Data
1		94

Write Buffer

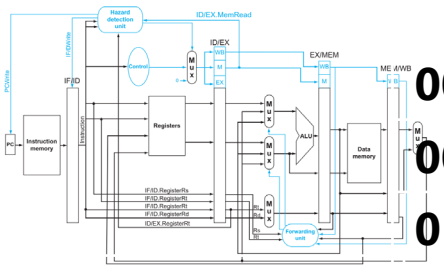
V	D	Tag	Data
0	0		
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Write-back: Writes

- What happens if we write **94** to address **000010**$_2$? (cont'd)
  - Now we can finally read in block **93** to the L1 Cache



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	0	0		

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93

Memory

Address	Data
...	...
000010	93
...	...
111010	24
...	...

Store Queue

V	Tag	Data
1		94

Write Buffer

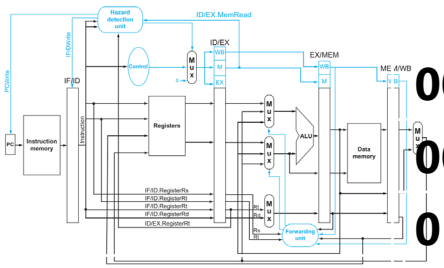
V	D	Tag	Data
1	1	111	25
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Write-back: Writes

- What happens if we write **94** to address **000010**$_2$? (cont'd)
  - Now we can finally read in block **93** to the L1 Cache
  - And write **94** into the cache block, also marking it dirty
  - Store is finished, so now remove it from the Store Queue!



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	000	94

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93

Memory

Address	Data
...	...
000010	93
...	...
111010	24
...	...

Store Queue

V	Tag	Data
1		94

Write Buffer

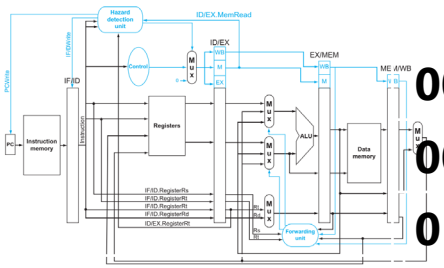
V	D	Tag	Data
1	1	111	25
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Write-back: Writes

- What happens if we write **94** to address **000010**$_2$? (cont'd)
  - Eventually, the pending block in L1 Write Buffer will write back
  - But this didn't affect the original store latency



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	000	94

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93

Memory

Address	Data
...	...
000010	93
...	...
111010	24
...	...

Store Queue

V	Tag	Data

Write Buffer

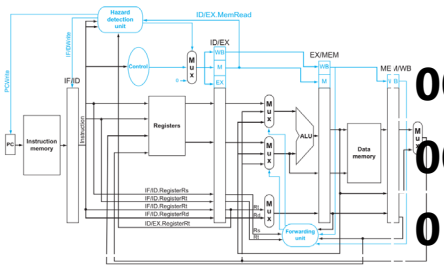
V	D	Tag	Data
1	1	111	25
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Write-back: Reads

- What happens if we read **25** from address **111010**$_2$?
  - Misses in L1 and L2 caches and must go all the way to Memory



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	000	94

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	000	93

Memory

Address	Data
...	...
000010	93
...	...
111010	25
...	...

Store Queue

V	Tag	Data

Write Buffer

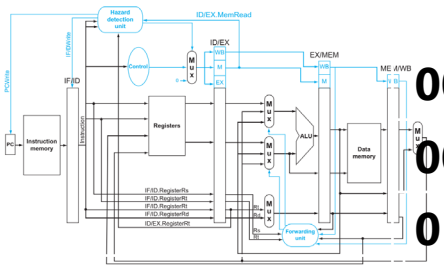
V	D	Tag	Data
0	0		
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Write-back: Reads

- What happens if we read **25** from address **111010**$_2$?
  - Misses in L1 and L2 caches and must go all the way to Memory
  - Fills the L2 Cache with **25** on the way back after evicting block **93** (Note that block **93** can simply be discarded since it's clean)



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	000	94

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	111	25

Memory

Address	Data
...	...
000010	93
...	...
111010	25
...	...

Store Queue

V	Tag	Data

Write Buffer

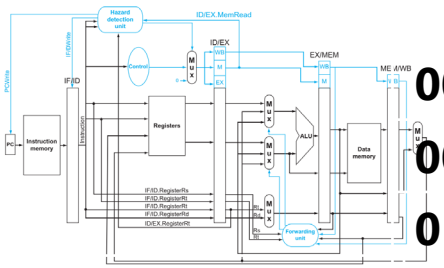
V	D	Tag	Data
0	0		
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Write-back: Reads

- What happens if we read **25** from address **111010**$_2$? (cont'd)
  - Now it needs to evict block **94** in L1 Cache before filling with **25**
  - But block **94** needs to be written back since it's dirty!
  - So move to Write Buffer temporarily to make space.



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	000	94

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	111	25

Memory

Address	Data
...	...
000010	93
...	...
111010	25
...	...

Store Queue

V	Tag	Data

Write Buffer

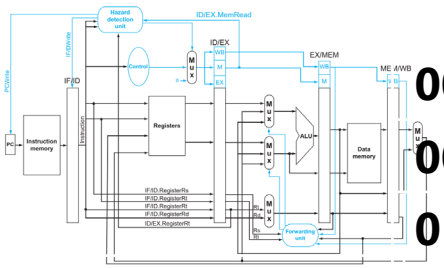
V	D	Tag	Data
0	0		
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Write-back: Reads

- What happens if we read **25** from address **111010**$_2$? (cont'd)
  - Now L1 Cache can be filled with block 25



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	0	0		

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	111	25

Memory

Address	Data
...	...
000010	93
...	...
111010	25
...	...

Store Queue

V	Tag	Data

Write Buffer

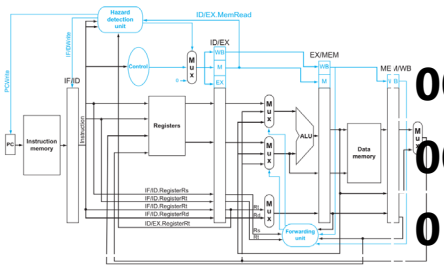
V	D	Tag	Data
1	1	000	94
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Write-back: Reads

- What happens if we read **25** from address **111010**$_2$? (cont'd)
  - Now L1 Cache can be filled with block **25**
  - Block **94** will eventually be written back to Memory
  - Write buffers in this context are also called **victim caches**



L1 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	0	111	25

L2 Cache

	V	D	Tag	Data
000	0	0		
001	0	0		
010	1	1	111	25

Memory

Address	Data
...	...
000010	93
...	...
111010	25
...	...

Store Queue

V	Tag	Data

Write Buffer

V	D	Tag	Data
1	1	000	94
0	0		

Write Buffer

V	D	Tag	Data
0	0		
0	0		

Impact of Write Policy on AMAT

- AMAT = hit time + (miss rate × miss penalty)

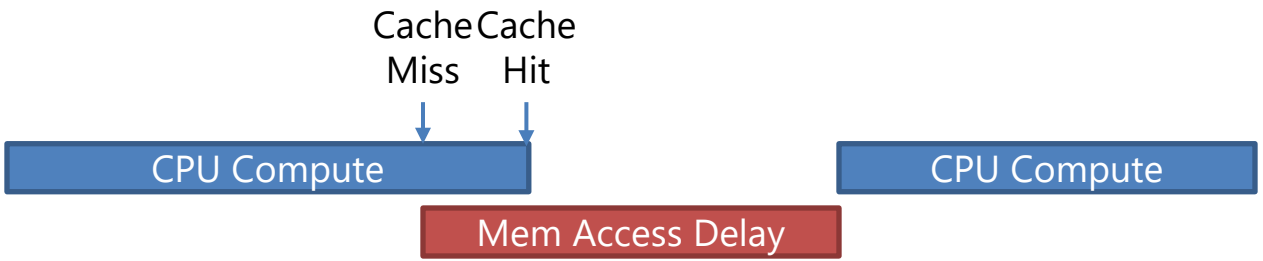
- **Write-through** caches have a **longer write hit time**
  - With write-back, a read hit and write hit take the same amount of time
  - With write-through, a write hit takes the same time as a write miss
  - Can be mitigated using **store queues**
- **Write-back** caches can have **longer miss penalty**
  - Due to write-backs of dirty blocks when making space for new block
  - Can be mitigated using **store queues** and **write buffers**

- All in all, write-back caches usually outperform write-through caches
  - Because write hits are typically more frequent compared to write-backs
- But write-through sometimes used in L1 cache due to simplicity
  - Plenty of L1 → L2 (intra-chip) bandwidth to handle write propagation
  - For L3, L3 → DRAM bandwidth cannot support write propagation

Cache Design Parameter 6: Blocking vs. Non-blocking

Blocking caches can only service one request at a time

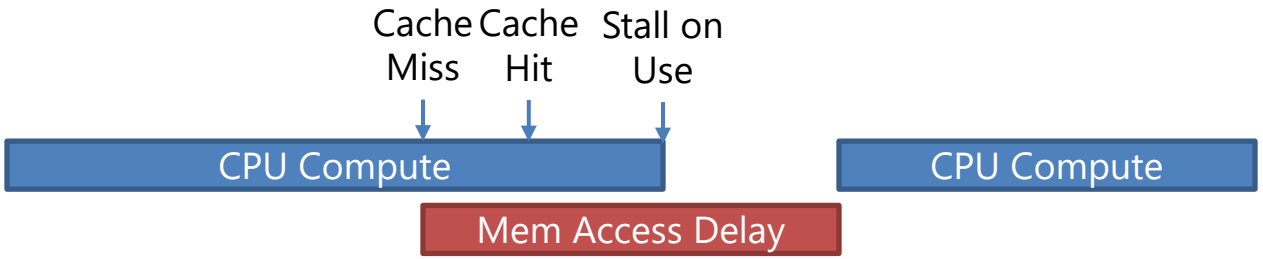
- Blocking caches:



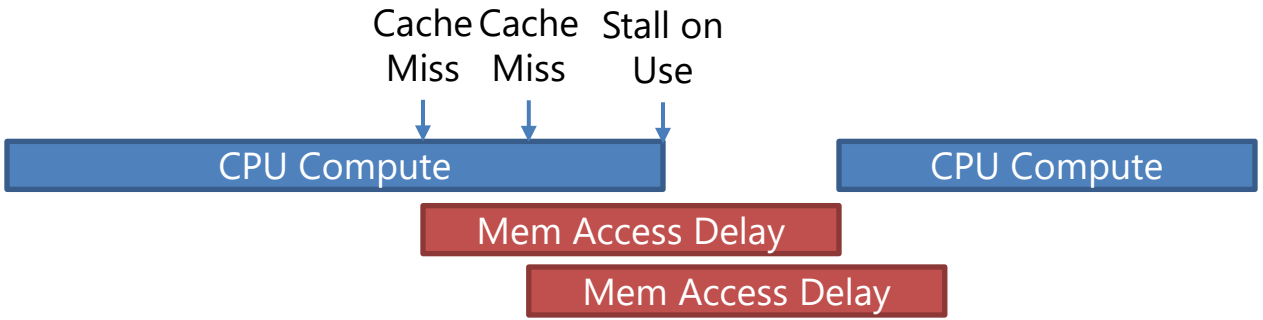
- After cache miss, cache goes into “not ready” state until it returns
- CPU can continue computation through out-of-order execution
- But, on second cache access, cache will not respond even on hit

Non-blocking caches service requests concurrently

- Hit under miss:

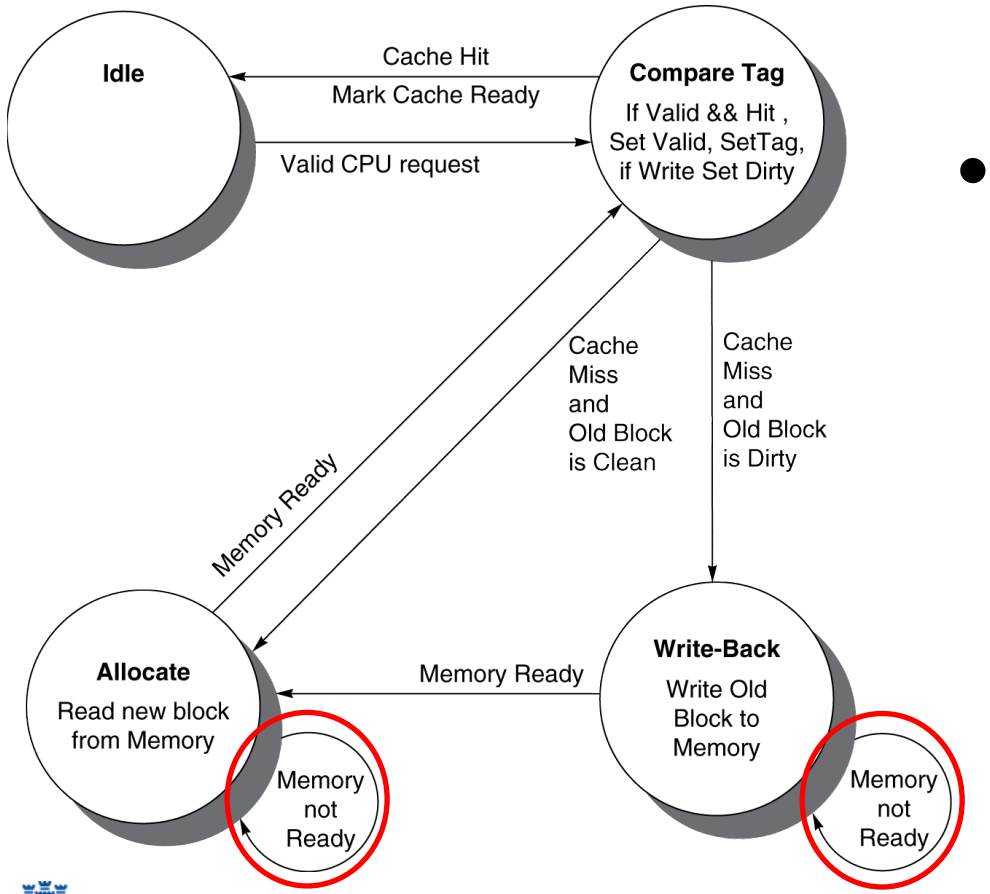


- Miss under miss:



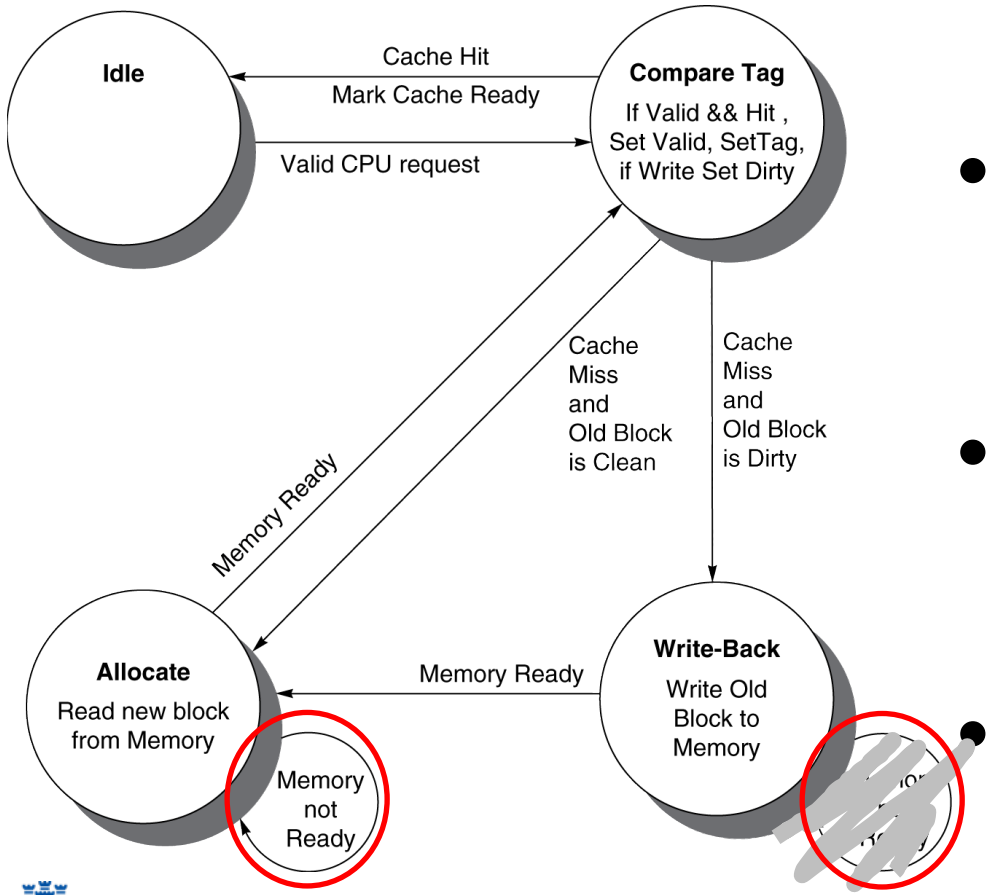
- **Non-blocking cache** allows both to happen
  - Allows **Memory Level Parallelism (MLP)**
  - As important to performance as Instruction Level Parallelism (ILP)

Blocking Cache FSM for Write Back Caches



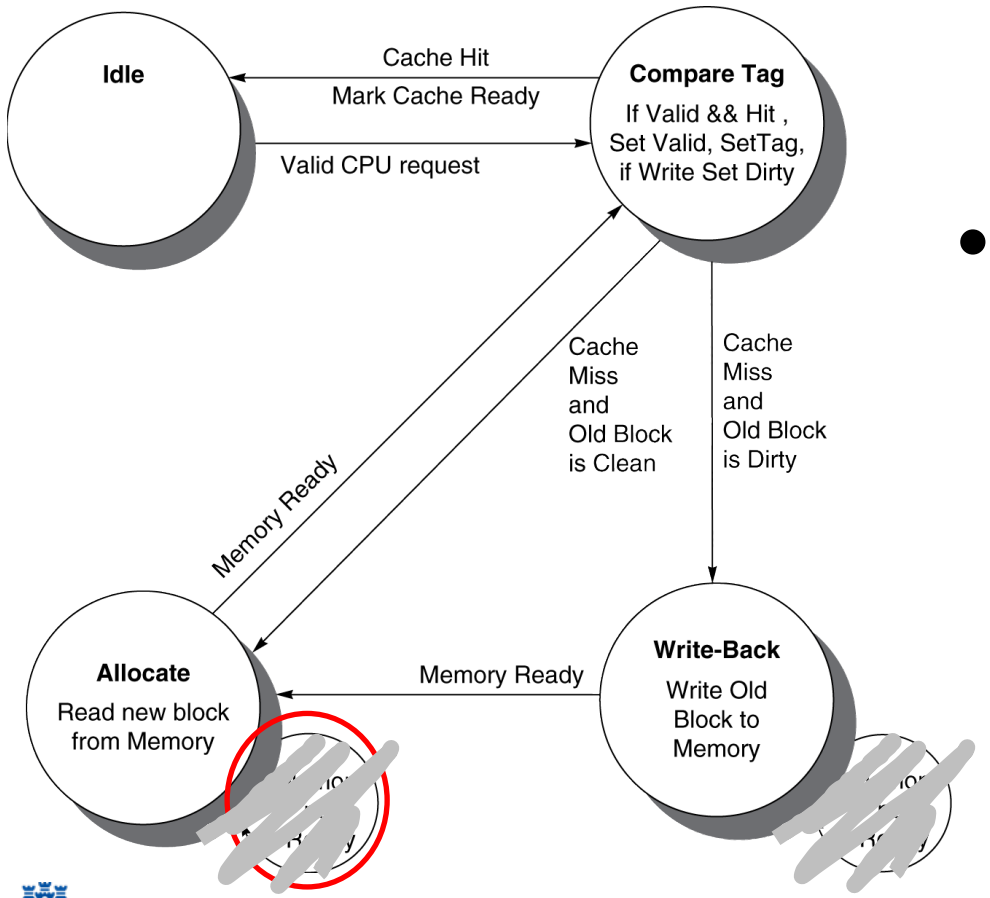
- FSM must be in **Idle** state for cache to receive new requests
- Prolonged "Memory not Ready" blocks quick return to idle state → Called **Blocking Cache**

Non-Blocking Cache FSM w/ Write Buffer



- FSM must be in **Idle** state for cache to receive new requests
- Prolonged "Memory not Ready" blocks quick return to idle state → Called **Blocking Cache**
- **Write buffer** allows cache to defer **write-back** until later
  - Quick transition to allocate
- But how about "Memory not Ready" on Allocate state?

Non-Blocking Cache FSM w/ Write Buffer + MSHR



- On cache miss:
  → Miss registered in MSHR.
  → Cache quickly returns to idle.

- **M**iss **S**tatus **H**andling **R**egister
  - A table that keeps track of pending missed requests.
  - When request returns,
    → Cache block filled with data.
    → Request forwarded up the memory hierarchy.

Impact of non-blocking caches

- Non-blocking caches do not impact our three cache metrics
  - Hit time, miss rate, and miss penalty remain mostly the same

- Impact is that **miss penalty** can be **overlapped** with:
  - Computation of instructions not dependent on the miss
  - Miss penalties of other memory requests

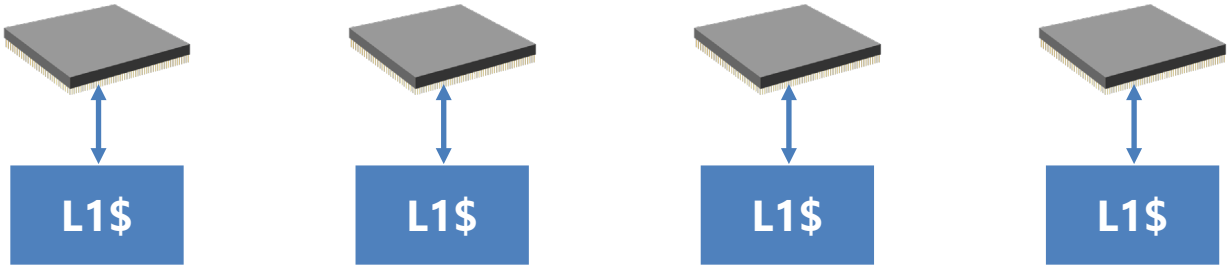
- Out-of-order processors are always coupled with a non-blocking cache
  - Otherwise, the ability to do out-of-order execution is severely stymied

Cache Design Parameter 7: Private vs. Shared

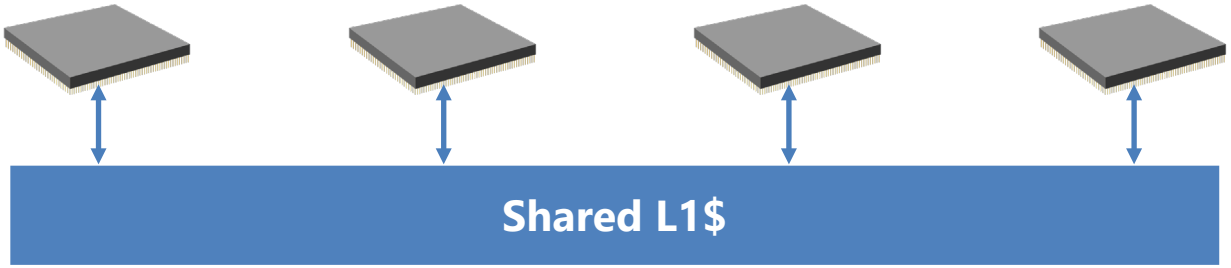
Private vs. Shared Cache

- On a multi-core system, there are two ways to organize the cache

- **Private** caches: each core (processor) uses its own cache

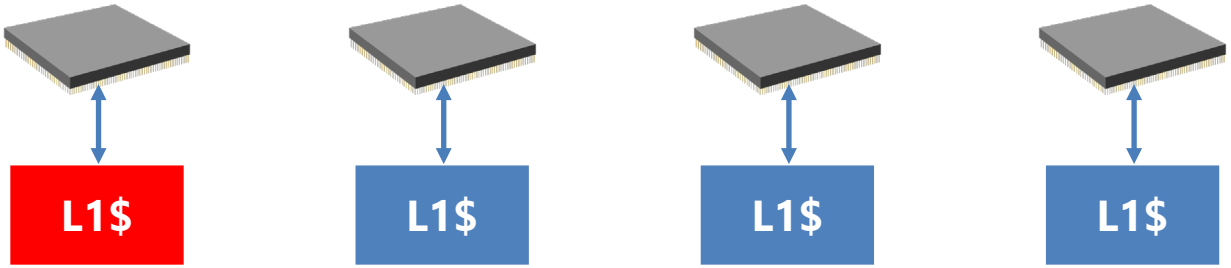


- **Shared** cache: all the cores share one big cache



Shared Cache can Use Space More Flexibly

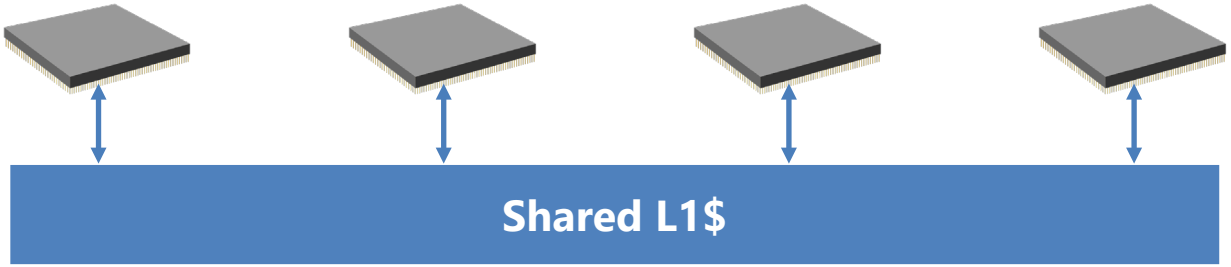
- Suppose only 1st core is active and other cores are idle
  - How much cache space is available to 1st core? (Shown in **red**)
- **Private** caches: 1st core can only use its own private cache



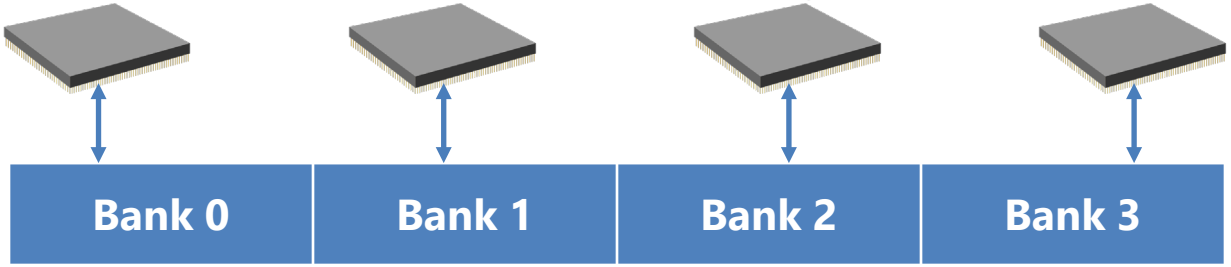
- **Shared** cache: 1st core can use entire shared cache!

Banking: Solution to Structural Hazards

- Now what if all the cores are active at the same time?
  - Won't that cause **structural hazards** due to simultaneous access?



  - Could add more ports, but adding **banks** is more cost effective

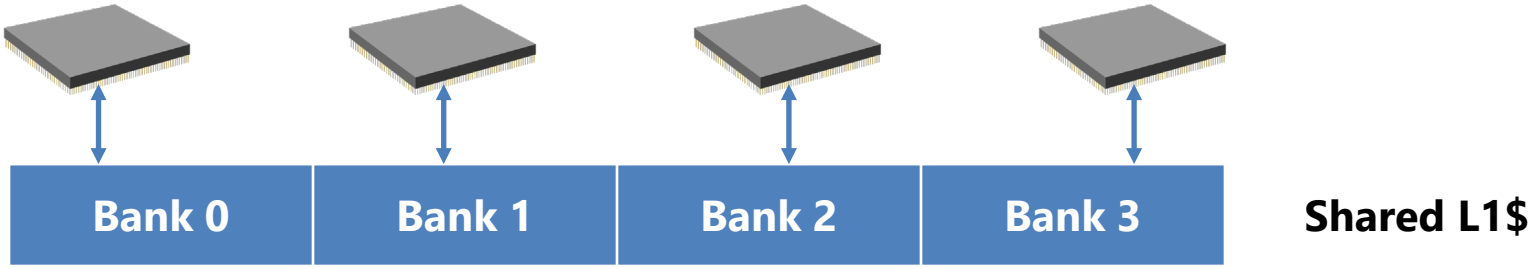


Shared L1$

    - Each bank has one read / write port
    - As long as two cores do not access same bank, no hazard!

Banking: Solution to Structural Hazards

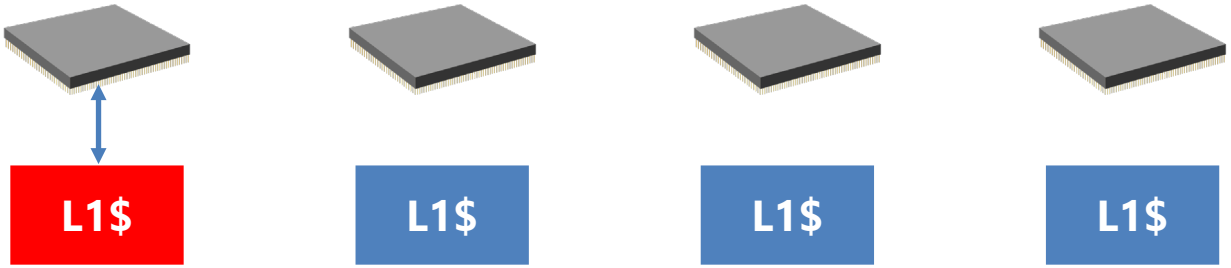
- Cache blocks are **interleaved** between banks



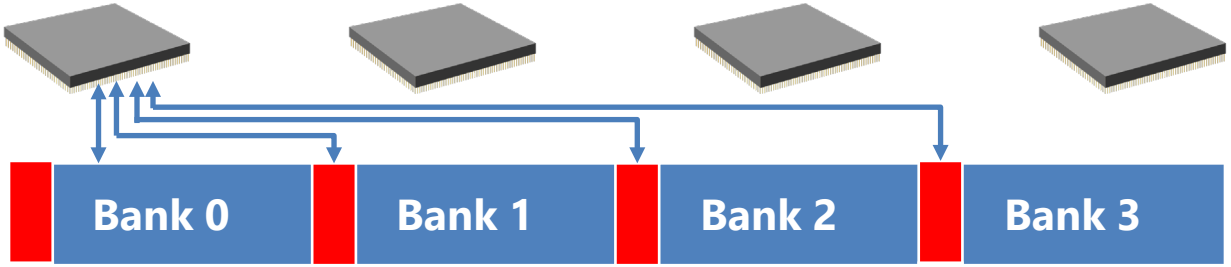
  - Blocks 0, 4, 8 … → Bank 0
  - Blocks 1, 5, 9 … → Bank 1
  - Blocks 2, 6, 10 … → Bank 2
  - Blocks 3, 7, 11 … → Bank 3
  - Why?
    - Spatial locality means cores likely to work on adjacent memory
    - Causes cache accesses to be evenly distributed → less hazards

Shared Cache have Longer Access Times

- Again, suppose only 1st core is active and other cores are idle
  - The working set data is shown in **red**
- **Private** caches: entire working set data in nearby private cache



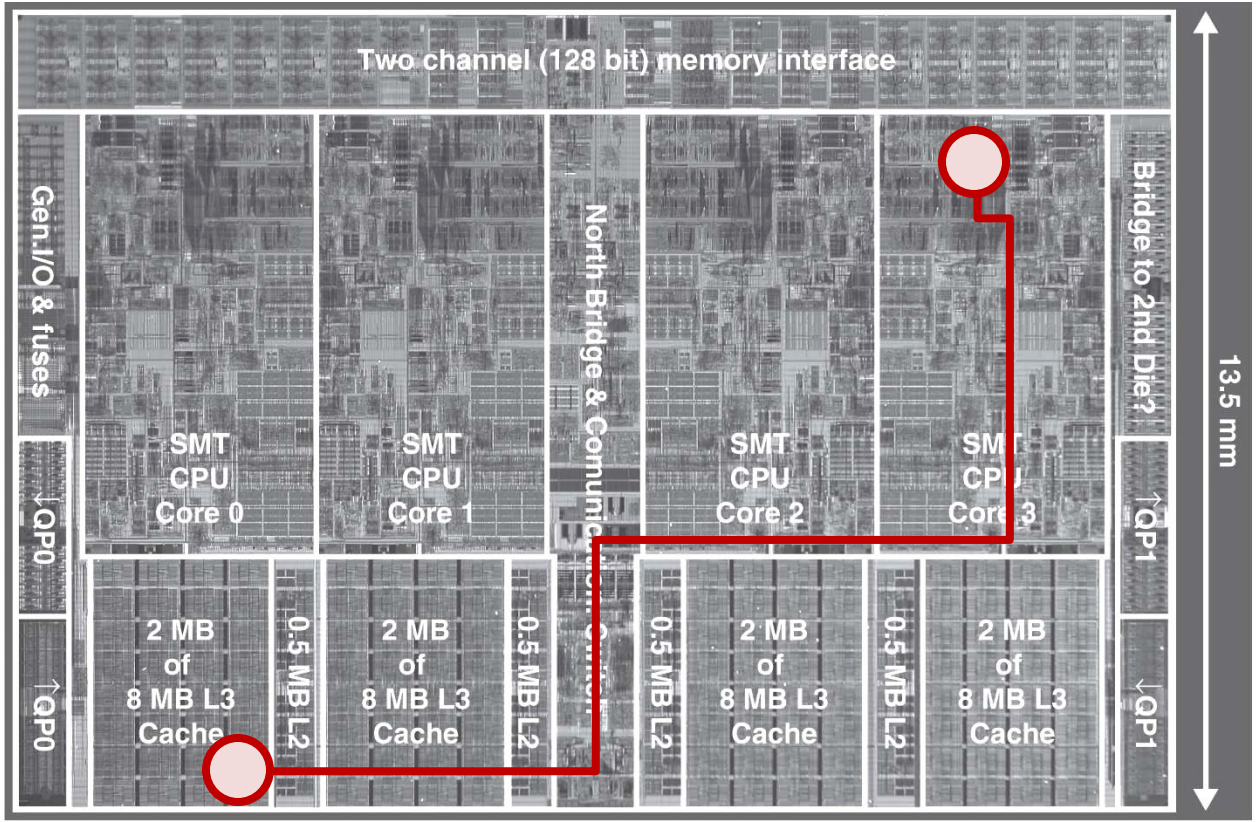
- **Shared** cache: data likely distributed to remote banks



Shared L1$

Shared Cache have Longer Access Times

- Remember this picture?



Impact of Shared Cache

- AMAT = hit time + (miss rate × miss penalty)

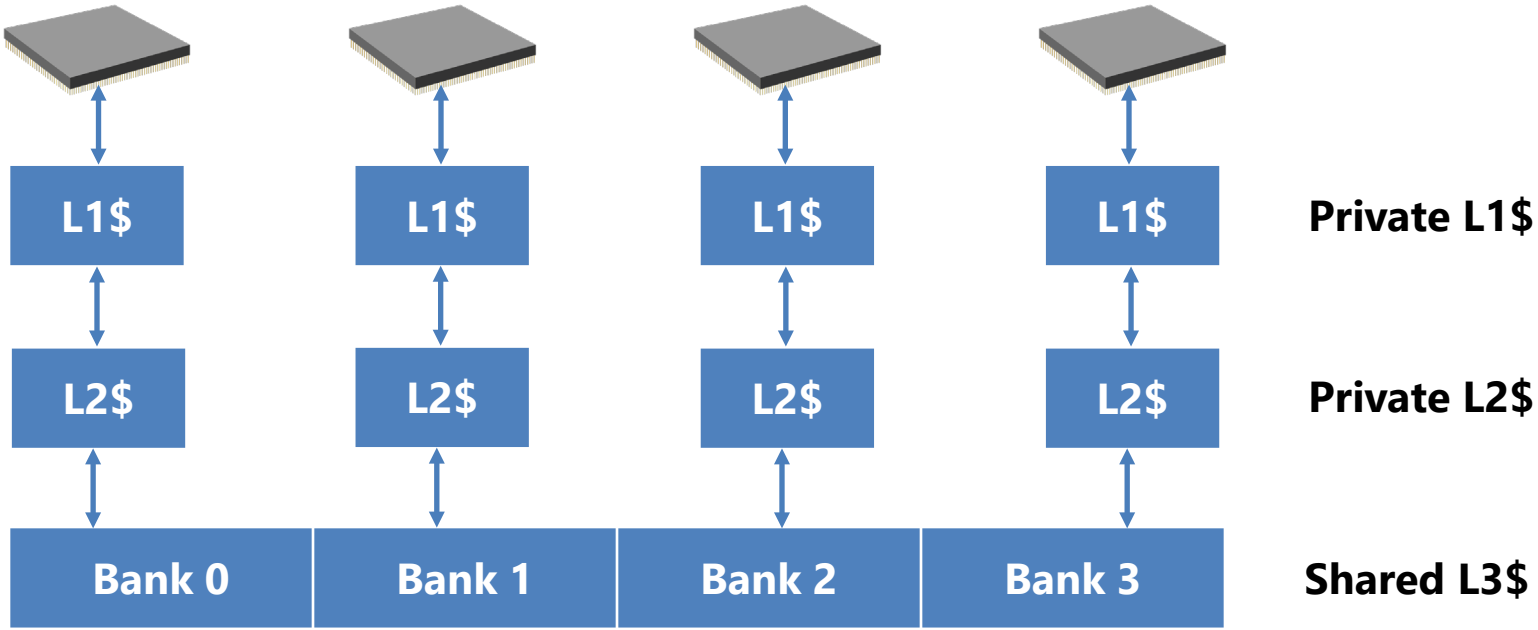
- Impact of shared cache on miss rate:
  - **Smaller miss rate** due to more flexible use of space

- Impact of shared cache on hit time:
  - **Longer hit time** due to sometimes having to access remote banks
  - **Longer hit time** due to port conflicts if multiple cores access same bank

- **L1** caches are almost always **private**
  - Hit time is important for L1. Cannot afford access to remote banks.
- **L3** (**last level**) caches are almost always **shared**
  - Reducing miss rate is top priority to avoid DRAM access.

Cache Organization of Broadwell CPU

- Cache organization of Skylake CPU used in thompson.cs.pitt.edu



- Intel rebrands the shared cache as the "Smart Cache"

Cache Design Parameter 8: Prefetching

Prefetching

- **Prefetching**: fetching data that is expected to be needed soon
  - Allows you to hide the latency of fetching that data
  - E.g. Web browsers prefetch resources from not-yet-clicked links → when user later clicks on link, response is almost instantaneous
  - Caches also prefetch data that is expected to be used soon
    - Can be used to avoid even **cold misses**

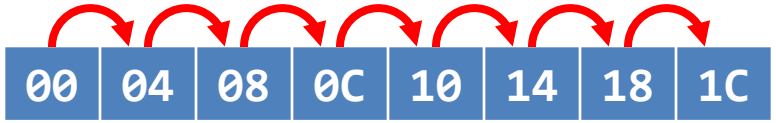
- Two ways prefetching can happen:
  - Compiler-driven: compiler emits **prefetch instructions**
    - Can manually insert one in C program: __builtin_prefetch(addr)
    - Or rely on compiler to insert them using heuristics
  - Hardware-driven: CPU **prefetcher** emits prefetches dynamically
    - Relies on prefetcher to detect a pattern in memory accesses

Hardware Prefetching

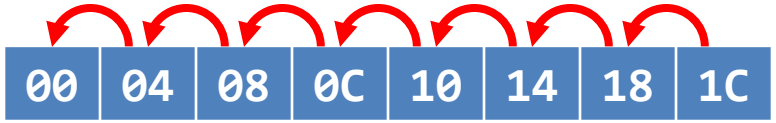
- What do you notice about both these snippets of code?
- They both access memory **sequentially.**
  - The first one data, the next instructions.
- These kinds of access patterns are very common.

```
for(i = 0 .. 100000)
    A[i]++;
```

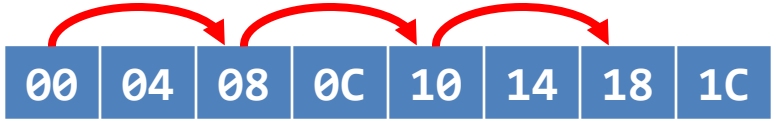
```
00 lw
04 lw
08 lw
0C addi
10 sub
14 mul
18 sw
1C sw
20 sw
```



Sequential



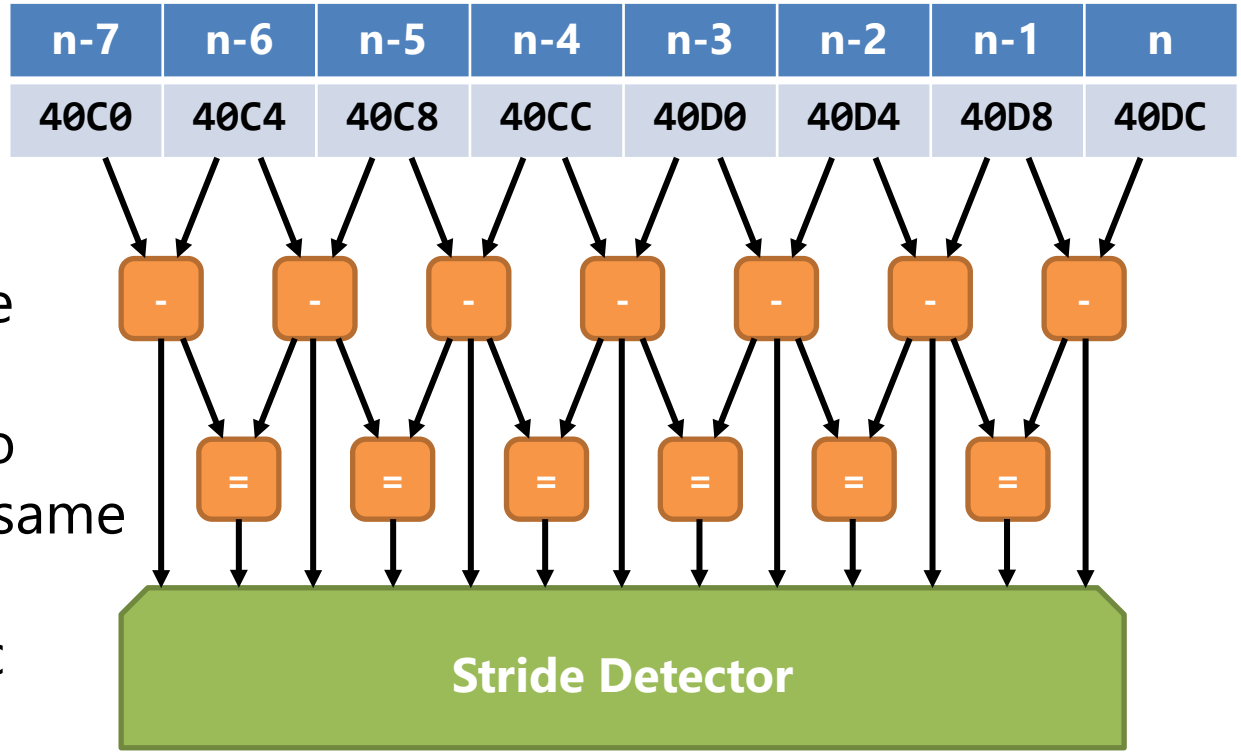
Reverse sequential



Strided sequential (think "accessing one field from each item in an array of structs")

Hardware Prefetching Stride Detection

- What kinds of things would you need?
- A table of the last *n* memory accesses would be a good start.

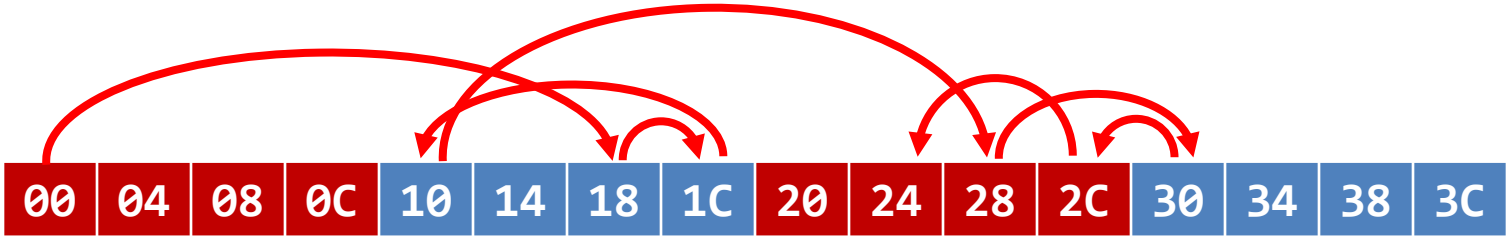


- Some subtractors to calculate the stride
- Some comparators to see if strides are the same
- Some detection logic

Where Hardware Prefetching Doesn't Work

- **Sequential** accesses are where prefetcher works best
  - E.g. Iterating over elements of an **array**

- Some accesses don't have a pattern or is too **complex** to detect
  - At below is how a typical **linked-list** traversal looks like



(colors are different cache blocks)

  - Other pointer-chasing data structures (**graphs**, **trees**) look similar
  - Can only rely on naturally occurring locality to avoid misses
  - Or, have compiler insert prefetch instructions in middle of traversal

Impact of Prefetching

- Prefetcher runs **in parallel** with the rest of the cache hardware
  - Does not slow down any on-demand reads or writes

- What if prefetcher is wrong? It can be wrong in two ways:
  - It fetched a block that was never going to be used
  - It fetched a useful block but fetched it too soon or too late
    - Too soon: the block gets evicted before it can be used
    - Too late: the prefetch doesn't happen in time for the access

- A bad prefetch results in **cache pollution**
  - Unused data is fetched, potentially pushing out other useful data
- On the other hand, good prefetches can reduce misses drastically!